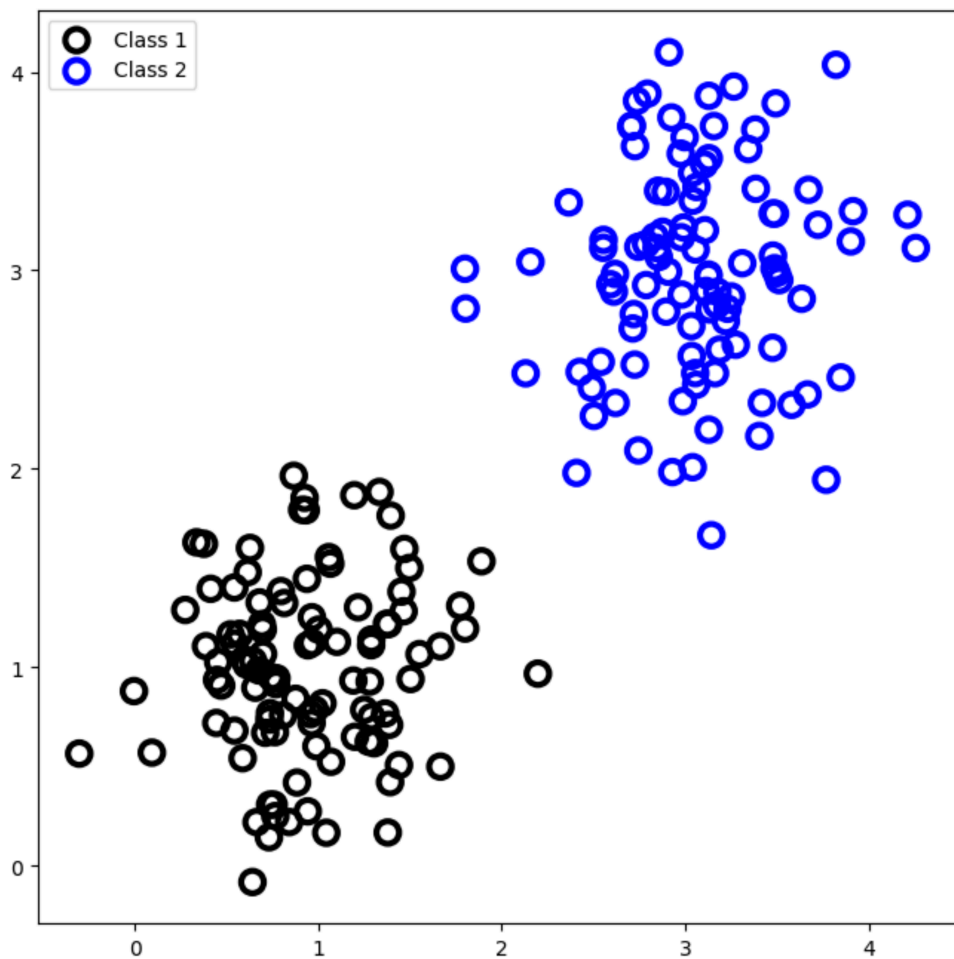


Support Vector Machine (SVM)

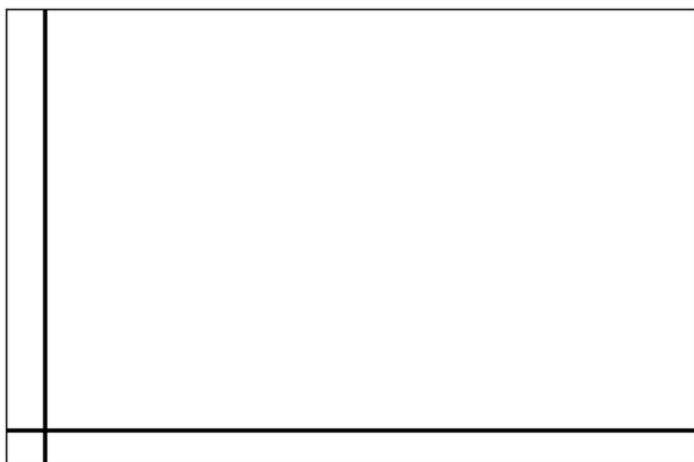
Introduction

- Last week we finished the lecture by learning about the Perceptron.
- Iterative algorithm that will become very useful in the next lectures.
- But, also some serious limitations:
 - No training error -> assumes linear separability.
 - Not unique $\mathbf{w}$.
- Consider the following example:



Support Vector Machine (SVM)

- Find $g(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0$ such that the **margin** m_{svm} is maximum (optimal).
 - Leaves room for variation between training data and test data.
 - **Generalizes** better.



Support Vector Machine (SVM)

- Note: $g(\mathbf{x}')$ not unique: $\mathbf{w}' = a\mathbf{w}$
- Define:
$$g(\mathbf{x}') = \pm 1$$

for $\mathbf{x}$ on margin.
- Hence: $m_{\text{svm}} = \frac{2}{\|\mathbf{w}\|}$

SVM optimization problem

$$\min_{\mathbf{w}, w_0} J(\mathbf{w}, w_0) := \frac{1}{2} \|\mathbf{w}\|^2$$

$$\text{subject to } y_i (\mathbf{w}^T \mathbf{x}_i + w_0) \geq 1$$

- "A training algorithm for optimal margin classifiers" - (Boser, Guyon, Vapnik, 1992.)
- "Support-vector networks" - (Cortes and Vapnik, 1995.)
- How to optimize the objective function?
 - Lagrange multipliers!

Hint on Lagrangian optimization

- $\min_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$
- subject to
$$f(\boldsymbol{\theta}) = \theta_1 \theta_2$$
$$-3 \geq 0$$
- Primal problem $\boldsymbol{\theta}$.
- Note: At $\boldsymbol{\theta} = \boldsymbol{\theta}_* \Rightarrow$
$$\frac{\partial}{\partial \boldsymbol{\theta}} J(\boldsymbol{\theta})$$
$$= \lambda \frac{\partial}{\partial \boldsymbol{\theta}} f(\boldsymbol{\theta}),$$
for $\lambda > 0$

Defining the Lagrange function

- $L(\boldsymbol{\theta},$
 $\lambda)$
 $= J$
 $(\boldsymbol{\theta})$
 $- \lambda f$
 $(\boldsymbol{\theta})$

- Then:
 $\min_{\boldsymbol{\theta}}$
 L
 $(\boldsymbol{\theta},$

Illustrating the Lagrangian

- $\min_{\boldsymbol{\theta}} J(\boldsymbol{\theta})$
- subject to $f_i(\boldsymbol{\theta}) \geq 0$ for $i = 1, \dots, N$
- $\frac{\partial}{\partial \lambda} L$
 $(\boldsymbol{\theta})$
 $- \lambda \frac{\partial}{\partial \boldsymbol{\theta}} f$
 $(\boldsymbol{\theta})$
 $= 0$

Illustrating the Lagrangian

- Note: Only one solution $\boldsymbol{\theta}$.
- Due to $J(\boldsymbol{\theta})$ being convex and feasible region being a convex set.
- Lagrange function: $L(\boldsymbol{\theta}, \lambda) = J(\boldsymbol{\theta}) - \sum_{i=1}^N \lambda_i f_i(\boldsymbol{\theta})$
- Then: $\frac{\partial}{\partial \boldsymbol{\theta}} L(\boldsymbol{\theta}, \lambda) = \frac{\partial}{\partial \boldsymbol{\theta}} J(\boldsymbol{\theta}) - \sum_{i=1}^N \lambda_i \frac{\partial}{\partial \boldsymbol{\theta}} f_i(\boldsymbol{\theta}) = \mathbf{0}$

Illustrating the Lagrangian

- Note: only one **active** constraint: $f_1(\boldsymbol{\theta}) = 0$
- $\Rightarrow \lambda_1 \geq 0, \lambda_2 = 0, \lambda_3 = 0$
- Draw $\Rightarrow$

Karush-Kuhn-Tucker (KKT)

- The SVM optimization objective is a nonlinear optimization task subject to a set of linear constraints.
- Difficult to optimize, how do we know if a solution is optimal?
- The Karush-Kuhn-Tucker conditions is a well-known set of conditions for optimality in nonlinear optimization.
- At $\boldsymbol{\theta} = \boldsymbol{\theta}_*$ the KKT conditions read:
 - $\frac{\partial}{\partial \boldsymbol{\theta}} L(\boldsymbol{\theta}, \lambda) = \mathbf{0}$
 - $\lambda_i f_i(\boldsymbol{\theta}) = 0$
 - $\lambda_i \geq 0$

Dual problem (λ)

- Note:

$$\max_{\lambda \geq 0} L(\boldsymbol{\theta}_*, \lambda)$$

$$= J(\boldsymbol{\theta}_*) -$$

$$\sum_{i=1}^N \lambda_i f_i(\boldsymbol{\theta}_*)$$

- actually forces $\lambda_i f_i(\boldsymbol{\theta}_*) = 0$ since $\lambda_i \geq 0$ and $f_i(\boldsymbol{\theta}_*) \geq 0$.
- Hence:

$$\max_{\lambda \geq 0} L(\boldsymbol{\theta}_*, \lambda)$$

$$= J(\boldsymbol{\theta}_*)$$

Primal versus dual problem

- Putting it all together:

SVM Lagrangian

The Lagrangian for the SVM optimization problem is:

$$L(\mathbf{w}, w_0, \lambda) = \frac{1}{2} \|\mathbf{w}\|^2 - \sum_{i=1}^N \lambda_i [y_i(\mathbf{w}^T \mathbf{x}_i + w_0) - 1]$$

Taking the derivative with respect to w_0 :

$$\frac{\partial L}{\partial w_0} = - \sum_{i=1}^N \lambda_i y_i = 0$$

Taking the derivative with respect to $\mathbf{w}$:

$$\frac{\partial L}{\partial \mathbf{w}} = \mathbf{w} - \sum_{i=1}^N \lambda_i y_i \mathbf{x}_i = 0$$

Constraint KKT:

$$\sum_{i=1}^N \lambda_i y_i = 0$$

and

$$\lambda_i [y_i(\mathbf{w}^T \mathbf{x}_i + w_0) - 1] = 0$$

Remarks

- Let
$$y_i = 1$$
if
$$\lambda_i \neq 0$$
:
$$\mathbf{w}^T \mathbf{x}_i + w_0 = 1$$

Support vectors

- Those $\mathbf{x}_i$ that are subject to $\lambda_i \neq 0$ are on the ± 1 **margin**.
- Only $\mathbf{x}_i$ in the set of support vectors (SV) are important.
- Hence: $\mathbf{w} = \sum_{\mathbf{x}_i \in \text{SV}} \lambda_i y_i \mathbf{x}_i$
- And for the bias: w_0

Dual problem

Remember: $L(\mathbf{w}, w_0, \lambda) = \frac{1}{2} \|\mathbf{w}\|^2 - \sum_{i=1}^N \lambda_i [y_i(\mathbf{w}^T \mathbf{x}_i + w_0) - 1]$

Remarks

- Optimization is a science in itself (quadratic programming).
- Sequential minimal optimization (SMO, Platt)
- Dual problem: feature vectors enter loss in the form of an inner product.

Non-separable classes

- Allow errors in training $\Rightarrow$

- SVM objective:

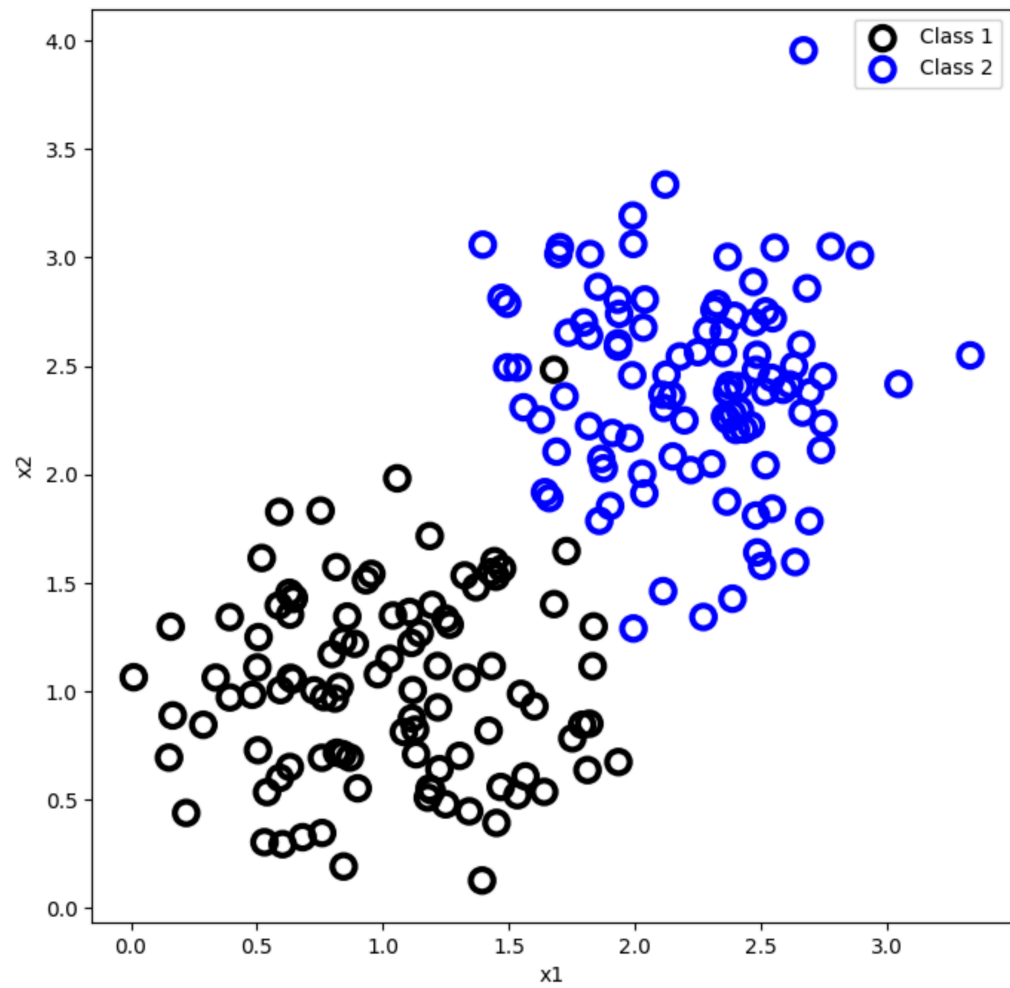
$$\min_{\mathbf{w}, w_0} J(\mathbf{w}, w_0)$$

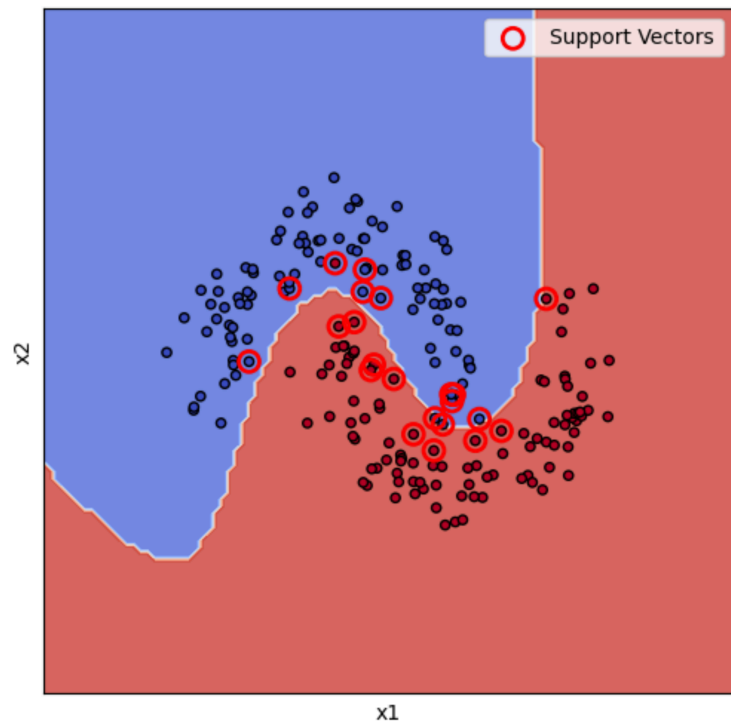
$$:= \frac{1}{2} \|\mathbf{w}\|^2$$

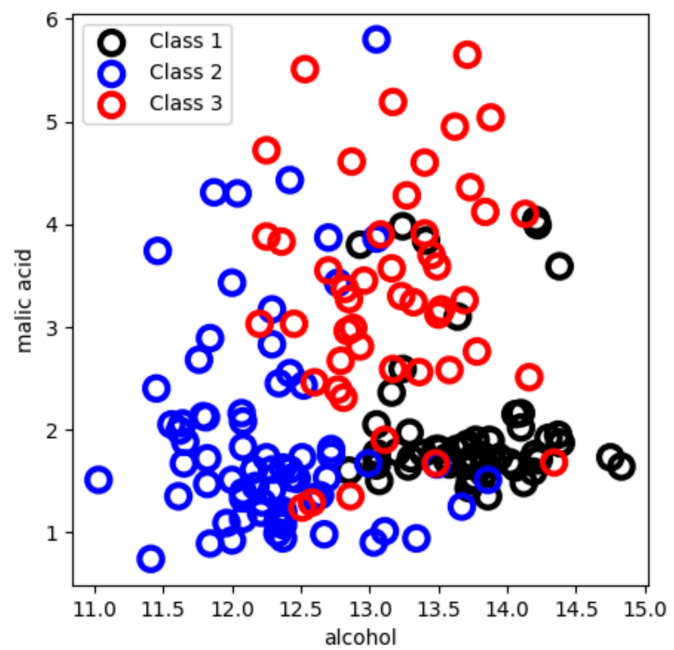
- subject to y_i

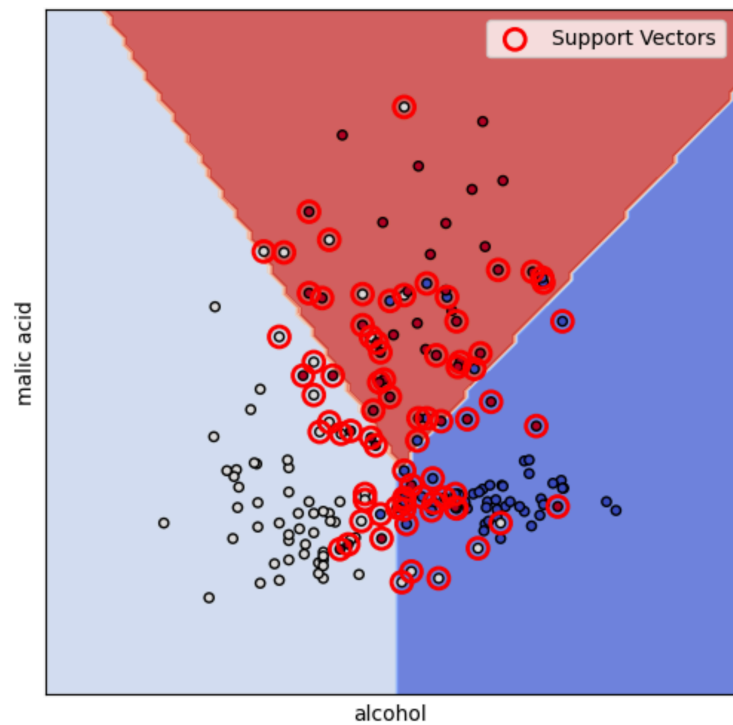
$$(\mathbf{w}^T \mathbf{x}_i + w_0) \geq 1$$

- Dual formulation:









Interactive SVM demo

Dataset Overlapping Gaussians (soft margin) ▼

Kernel Linear ▼

C (slack regularization) on $\log_{10}$ scale 

